Expert Systems With Applications 238 (2024) 121708

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

A stock series prediction model based on variational mode decomposition and dual-channel attention network

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ARTICLE INFO

Keywords:
Stock series prediction
Variational mode decomposition
Dual-channel attention network

ABSTRACT

Due to the comprehensive impact of external factors (politics, economy, market, etc.) and internal factors (organizational structure, management ability, innovation capability, etc.), stock series exhibit strong volatility. Coupled with their inherent high liquidity, it poses great challenges for stock series prediction. However, the previous stock series forecasting methods often only pay attention to the long-term dependencies, and lack attention to the local features and short-term dependencies. In this regard a stock series prediction model based on variational mode decomposition and dual-channel attention network is proposed, which is called VMD-LSTMA+TCNA. To prevent information leakage, the stock series is divided into equal-length sub-windows by sliding window. To reduce the series volatility, each sub-window is decomposed into different frequency mode sub-windows through variational mode decomposition (VMD). To improve the prediction accuracy and robustness in different stock markets, we construct a dual-channel attention model called LSTMA+TCNA. The LSTMA channel is used to extract long-term dependencies and temporally correlated features, while the TCNA channel is used to extract local patterns and short-term dependencies, and self-attention is added to both channels to increase the weight of features at important times. Predict each frequency mode sub-window separately through the specific LSTMA and TCNA channels, and then obtain the predicted values by fusing the results of dual-channel. The final predicted stock series is obtained by superimposing the predicted values of each frequency mode sub-window. Through extensive experiments on the US and Hong Kong stock markets, it has been shown that the VMD-LSTMA+TCNA model exhibits better robustness and generalization compared to other state-of-the-art methods and has higher prediction accuracy.

1. Introduction

One of the most concerning issues for practitioners in the stock trading market is the stock series change trend. Making reasonable and accurate predictions about stock series changes is a skill that practitioners must master. Due to the comprehensive impact of external factors (politics, economy, market, etc.) and internal factors (organizational structure, management ability, innovation capability, etc.), stock series exhibit strong volatility, such as non-stationary and irregularity (Lu et al., 2021). Therefore, stock series prediction is an attractive topic for researchers (De Fortuny et al., 2014). Based on the different principles of prediction, the existing methods can be divided into the following three categories: traditional machine learning methods, deep learning methods and hybrid prediction methods based on decomposition.

Traditional machine learning methods construct stock series prediction models through experience or optimization of small datasets, such as autoregressive moving average (ARMA), autoregressive integrated moving average (ARIMA), support vector machine (SVM) and other models (Adebiyi et al., 2014; Ariyo et al., 2014; Ince & Trafalis, 2008; Mintarya et al., 2023; Schierholt & Dagli, 1996; Zhao et al., 2021). However, the stock market is a complex system with many influencing factors, each affecting the other. The stock series has high liquidity and volatility, making capturing its characteristics with a specific model difficult. This is the main reason traditional methods are ineffective in predicting stock series.

In order to deal with the shortcomings of traditional methods in predicting stock series, deep learning technology has attracted more and more attention (Hu et al., 2021; Zhao et al., 2022). Chen et al.

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https://doi.org/10.1016/j.eswa.2023.121708
Received 17 May 2023; Received in revised form 24 August 2023; Accepted 16 September 2023
Available online 27 September 2023
0957-4174/

(2022), Ma et al. (2022), Roondiwala et al. (2017), Song et al. (2023) and Sunny et al. (2020) predict stock series using long short-term memory (LSTM) model. Li et al. (2019), Lu et al. (2020) and Selvin et al. (2017) combine the respective advantages of CNN and LSTM, then use CNN-LSTM to predict stock series. Variants of CNN, such as dilated causal convolutions, have shown great potential in time series tasks (Gan et al., 2021; Kok et al., 2020; Yuan et al., 2023; Zhao et al., 2023; Zhu et al., 2020). The self-attention mechanism was proposed by Vaswani et al. (2017) and has been widely used in the field of natural language processing (NLP) and computer vision (CV). Cheng et al. (2018), Liu et al. (2023) and Yang et al. (2018) introduced self-attention to improve prediction performance in stock series prediction. Although deep learning models are good at dealing with nonlinear and irregular data, they cannot fully identify and extract internal features from complex nonlinear and non-stationary data, especially in stock series prediction.

Although stock series prediction based on deep learning has achieved great success, it is difficult to capture the features of the original series directly. The decomposition strategy decomposes the original series into different frequency components, each corresponding to different features and having certain physical meanings. Compared with the black-box model of deep learning, the decomposition strategy is easier to explain the model's prediction results and is more conducive for decision-makers to understand and take action. Based on the hybrid method of decomposition, the original stock series is decomposed into multiple relatively stable subsequences by using the decomposition strategy, and then the subsequences are modeled separately for prediction, and finally, the prediction results are ensembled. It can be expressed as a “decomposition-prediction-ensemble”. Empirical mode decomposition (EMD) is a new adaptive signal time–frequency processing method, especially suitable for processing nonlinear and non-stationary series. Rezaei et al. (2021), Wei (2016), Yujun et al. (2020) and Zhou et al. (2019) employed the decomposition strategy of EMD and its variants to complement deep learning models for stock series prediction. However, EMD's multiscale decomposition of stocks suffers from mode mixing, which limits the model's prediction accuracy. To overcome this phenomenon, Ensemble Empirical Mode Decomposition (EEMD) mitigates the mode mixing of EMD decomposition by adding pairs of positive and negative Gaussian white noises to decompose signals. Yang and Wang (2021) combination of EEMD and neural network is used to construct a hybrid prediction model, but the intrinsic mode functions obtained by decomposing the stock series with this algorithm always have a certain amount of white noise remaining, affecting the model prediction performance. Residual white noise, which affects the prediction performance of the model. Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) introduces an adaptive noise processing mechanism, which can better deal with non-stationary signals and signals containing noise, Lin et al. (2022) and Lv et al. (2022) through CEEMDAN to solve the mode mixing phenomenon of the EMD to improve the prediction performance of stock indices, but the drawback of CEEMDAN is that the EEMD and neural network are combined to construct a hybrid prediction model. The disadvantage of CEEMDAN is that the number of mode decompositions cannot be determined. On the other hand, Variational Mode Decomposition (VMD) is a new type of adaptive, completely non-recursive method for mode variational and signal processing, which can overcome the problem of mode mixing of the EMD method, and can reduce the non-stationarity of time series with high complexity and strong nonlinearity and at the same time, it has the advantage of being able to determine the number of mode decompositions, and thus it has already been used in more applications in time series forecasting (Lahmiri, 2016; Niu et al., 2020; Wang et al., 2019) combines VMD and LSTM to improve stock prediction performance over other single deep learning methods, however, its limitation is that the decomposition will cause the information of the test set to leak into the training set, so it is not usable in practice. Liu et al. (2022) solves the problem of information leakage by changing the decomposition object through the sliding window and finding the optimal initialization parameters of LSTM through meta-learning algorithms to improve the prediction accuracy of the stock series. However, it has too few learnable parameters to explore the deeper features of the stock, and these methods lack of focusing on the local features and short-term dependencies of the stock series, so there is still room for improvement in the prediction accuracy.

In order to effectively improve the stock series forecasting accuracy, this paper proposes a new hybrid stock series prediction model named VMD-LSTMA+TCNA. The model combines the VMD method and constructs the dual-channel attention network consisting of LSTMA and TCNA to extract the long-term and short-term dependencies in the stock series, which equips it with the ability to deal with the high volatility stock series and to improve the prediction accuracy of the stock series. The contributions of this paper are summarized as follows.

- To reduce the volatility of the stock series, it is decomposed into some approximately stationary series with different frequencies by the VMD method.
- In order to improve the prediction accuracy and robustness for different stock markets, the dual-channel network composed of the LSTMA channel to extract long-term dependencies and temporal correlation features, and the TCNA channel to extract local patterns and short-term dependencies features, can effectively improve the prediction accuracy of stock series.
- To improve the prediction accuracy, the self-attention mechanism is introduced into the dual-channel network to better capture the correlation of time series.
- Extensive experiments on real US and Hong Kong stock markets demonstrate that the proposed model outperforms other latest models regarding prediction accuracy.

2. Related work

2.1. Variational mode decomposition

Variational mode decomposition (VMD) is a novel adaptive and completely non-recursive signal processing method proposed by Dragomiretskiy and Zosso (2013). Compared to other decomposition methods, VMD not only can determine the number of mode decompositions but also has a more robust mathematical theory and can reduce the non-stationarity of time series with high complexity and strong nonlinearity.

To obtain the bandwidths of the multiple modes after the VMD method, the original signal, denoted as $f(t)$, can be decomposed into K intrinsic mode functions (IMFs), represented by $u_i(t), i = 1, 2, \ldots, K$.

$$\begin{aligned}&\min_{\{\mu_i\}\{\omega_i\}} \left\{ \sum_{i=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_i(t) \right] e^{-j\omega_i t} \right\|_2^2 \right\} \\ &s.t. \sum_{i=1}^{K} u_i(t) = f(t)\end{aligned} \tag{1}$$

where K represents the number of IMFs, $u_k(t)$ represents the set of the ith IMF, w_i represents the set of center frequencies corresponding to the ith mode, and $\delta(t)$ is the Dirac function, $*$ for convolutional operations. To solve this problem, the aforementioned optimization problem with equality constraints can be transformed into an unconstrained optimization problem by augmenting the Lagrangian function.

$$\begin{aligned}&L\left(\{u_i\}, \{\omega_i\}, \lambda\right) \\ &= \alpha \sum_{i=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_i(t) \right] e^{-j\omega_i t} \right\|_2^2 \\ &+ \left\| f(t) - \sum_{i=1}^{K} u_i(t) \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_{i=1}^{K} u_i(t) \right\rangle\end{aligned} \tag{2}$$

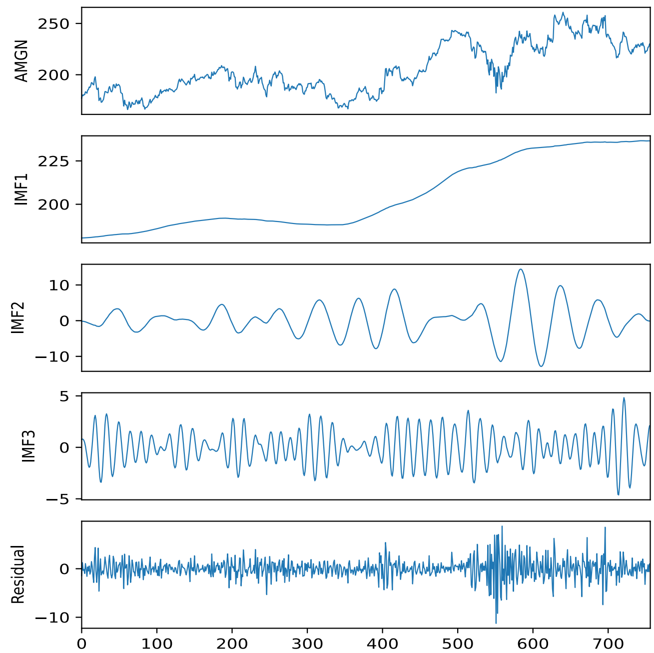

Fig. 1. The decomposition results of Amgen.

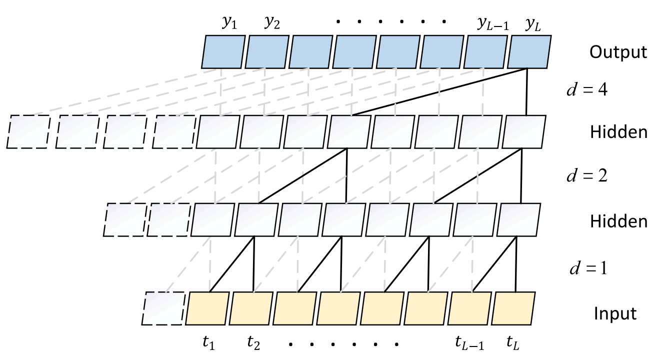

Fig. 2. Schematic diagram of dilated causal convolution.

where λ is the Lagrange multiplier and α is the quadratic penalty term. Solving the above problem solving using the alternating direction multiplication method (ADMM) yields $\{\mu_i\}$ and $\{\omega_i\}$.

The stock series of AMGEN in the DJIA stock dataset is decomposed into three subsequences and residual items. As depicted in Fig. 1, the subsequence obtained through the VMD method is more stable and regular, making it particularly suitable for noisy and unstable financial time series. Therefore the method proposed in this paper uses VMD to decompose the stock series to attenuate the high volatility and also compares VMD with other decompositions such as Wavelet Decomposition (DWT), EMD, EEMD, and CEEMDAN.

2.2. Dilated causal convolution

Convolutional neural network (CNN) is widely used in computer vision. When processing images, CNN treats the image as a two-dimensional matrix. When applied to time series data, the time series can be seen as a one-dimensional vector. Using CNN for time series prediction allows for parallel processing, but there are two issues with directly applying the CNN structure: information leakage in the hidden layer, and the size of the receptive field has a linear relationship with the depth of the network, which does not fully utilize the timing information.

Temporal convolutional network (TCN) applies causal convolution to solve the timing issue. As shown in Fig. 2, 0-padding ensures that the input and output lengths are consistent. At the same time, to preserve the timing of time series data, the output at time t will only use data at time t and before, which avoids information leakage and improves the generalization ability of the model.

Dilated convolution is introduced to increase the receptive field without losing partial information, and dilated convolution injects holes into the standard CNN to expand the receptive field. The combination of causal convolution and dilated convolution ensures the causal relationship in time. The number of holes increases as the network layer deepens, and more padding needs to be added to maintain the causal relationship. The dilatation rate controls the spacing between the values in the convolution kernel. The size of the dilation rate d is exponentially related to the depth of the network, so the size of the receptive field is exponentially related to the depth of the network. This allows the output of each convolution kernel to contain more information. The combination of causal and inflationary convolution ensures that the model does not introduce interference from future information when extracting features, which is extremely important in stock series forecasting tasks. It also retains sensitivity to local features while expanding the sensory field, which better captures local features and short-term dependencies of the stock series.

3. Methodology

To reduce the liquidity and volatility and improve the prediction accuracy of the stock series, this paper proposed a stock series prediction model based on variational mode decomposition and dual-channel attention network, which is called VMD-LSTMA+TCNA. As shown in Fig. 3, the stock series is first segmented into multiple sub-windows using sliding window to prevent information leakage. Then, these sub-windows are decomposed into many different frequency mode sub-windows by the VMD, which indirectly reduces volatility and increases the stability of the stock series. Next, dual-channel attention network consisting of LSTMA and TCNA is utilized to extract the long-term and short-term dependencies of stocks. During the backpropagation process, the dual-channel attention network can calculate appropriate weights for each channel, making the dual-channel complementary and thus improving the robustness and generalization of the model, with good performance in different stock trading markets. Finally, all of the predicted values of the sub-windows with different frequencies are summed together to obtain the final predicted stock series.

3.1. Sliding window and VMD

For the stock series $X = \{x_1, x_2, \ldots, x_N\}$, where x_i represents the closing price on the ith day, and N is the total number of days in stock series. Let sub-window W_i define as Eq. (3).

$$\begin{aligned} W &= \{W_1, W_2, \ldots, W_{N-L}\} \\ W_1 &= \{x_1, x_2, \ldots, x_L\} \\ W_2 &= \{x_2, x_3, \ldots, x_{L+1}\} \\ &\ldots\ldots \\ W_{N-L} &= \{x_{N-L}, x_{N-L+1}, \ldots, x_{N-1}\} \end{aligned} \tag{3}$$

where L is the length of the selected sub-window, representing the number of days. Therefore, a total of $(N - L + 1)$ sub-windows are generated, and all of them are combined to form the input dataset W.

It should be noted that the last value x_N of the last sub-window W_{N-L+1} corresponds to the closing price on the Nth day. But there is no closing price on day $(N + 1)$th to verify the accuracy of the predicted value, so the last sub-window W_{N-L+1} is discarded, and the input dataset W is formed by sub-windows $W_1, W_2, \ldots, W_{N-L}$.

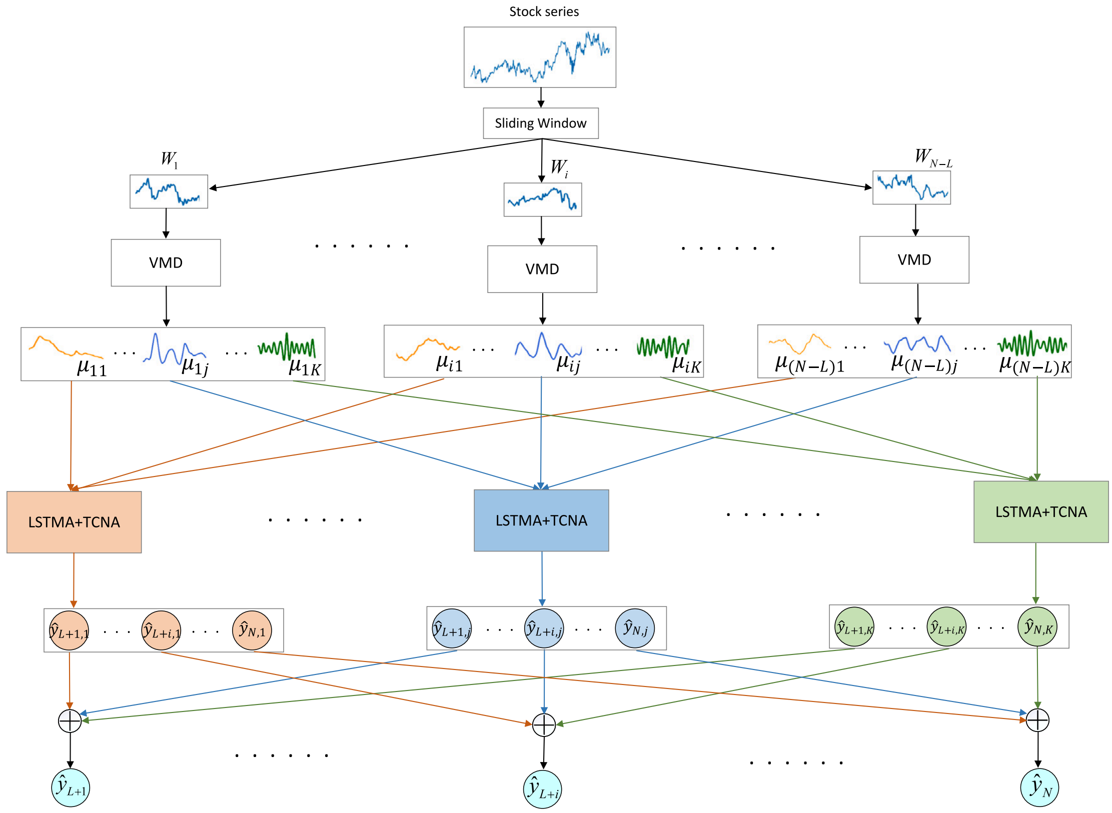

Fig. 3. Flow chart of the VMD-LSTMA+TCNA model.

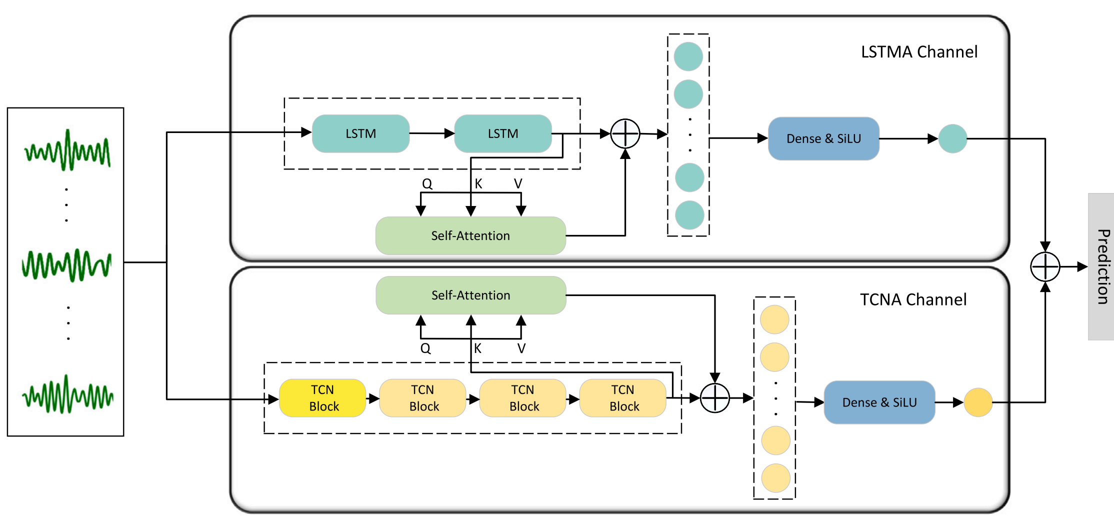

Fig. 4. The architecture of LSTMA+TCNA model for a specific frequency mode sub-window.

Therefore, we can predict the closing price $\hat{y}_{i+L}$ on day $(i+L)$th by the sub-window W_i:

$$\begin{aligned}\hat{y}_{i+L} &= F(W_i) \\ &= F(x_i, x_{i+1}, \ldots, x_{i+L-1})\end{aligned} \tag{4}$$

where $F(\cdot)$ is the VMD-LSTMA+TCNA model.

In order to effectively reduce noise interference and data volatility, each sub-window W_i is decomposed into K mode sub-windows of $\{\boldsymbol{\mu}_{i1}, \boldsymbol{\mu}_{i2}, \ldots, \boldsymbol{\mu}_{iK}\}$ with different frequencies by the VMD method defined as Eq. (1). Furthermore, they meet the following relationships:

$$\sum_{j=1}^{K} \boldsymbol{\mu}_{ij} = W_i \tag{5}$$

where K represents the number of decompositions in VMD.

Therefore, the problem in Eq. (4) can be transformed into predicting the K different frequency mode sub-windows, as shown below.

$$\hat{y}_{i+L} = \sum_{j=1}^{K} F'_j(\boldsymbol{\mu}_{ij}) = \sum_{j=1}^{K} \hat{y}_{i+L,j} \tag{6}$$

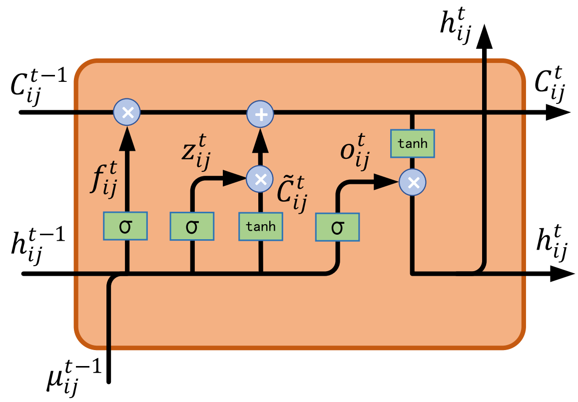

Fig. 5. LSTM unit structure.

where, $F'_j(\cdot)$ is the LSTMA+TCNA model.

3.2. LSTMA+TCNA

In order to improve the prediction accuracy and robustness in different stock markets, we construct a dual-channel attention model called LSTMA+TCNA. It comprises a self-attention long short-term memory network (LSTMA) and a self-attention temporal convolutional network (TCNA). The self-attention mechanism in the dual-channel attention network can effectively capture hidden features in the stock series while removing noise interference and improving the prediction accuracy.

The architecture of LSTMA+TCNA is shown in Fig. 4. The jth mode sub-window $\boldsymbol{\mu}_{ij}$ is predicted by the specific frequency $LSTMA_j$ and $TCNA_j$ model. Then, they are fused to obtain predicted values $\hat{y}_{i+L,j}$ for specific frequency j.

$$\hat{y}_{ij} = LSTMA_j(\boldsymbol{\mu}_{ij}) + TCNA_j(\boldsymbol{\mu}_{ij}) \tag{7}$$

3.2.1. LSTMA channel

As shown in the upper part of Fig. 4, LSTMA for extracting long-term dependencies and time features, which consists of two LSTM layers, a self-attention layer, and a "Dense&SiLU" module.

LSTM is used to overcome the problem of gradient vanishing or exploding in long sequence tasks of RNN (Hochreiter & Schmidhuber, 1997). The LSTM is the backbone network for this channel, which introduces memory cells and gating mechanisms to selectively retain and transfer information, allowing it to efficiently process sequence data over long time intervals. This mechanism makes it ideally suited for extracting global features and long-term dependencies of stock series. The key components of the LSTM unit include input gate, forget gate, output gate, and memory cell (Fig. 5). These gating mechanisms control the flow of information, allowing the network to capture and remember relevant information while discarding irrelevant or redundant information.

The ith sub-window of the jth mode $\boldsymbol{\mu}_{ij} = \{\mu_{ij}^1, \ldots, \mu_{ij}^t, \ldots, \mu_{ij}^L\}$, whose long-term dependencies is extracted by the following equation:

$$\begin{aligned}
z_{ij}^t &= \sigma\left(W_z \cdot \left[h_{ij}^{t-1}, \mu_{ij}^t\right] + b_z\right) \\
o_{ij}^t &= \sigma\left(W_o \cdot \left[h_{ij}^{t-1}, \mu_{ij}^t\right] + b_o\right) \\
f_{ij}^t &= \sigma\left(W_f \cdot \left[h_{ij}^{t-1}, \mu_{ij}^t\right] + b_f\right) \\
\tilde{C}_{ij}^t &= \tanh\left(W_c \cdot \left[h_{ij}^{t-1}, \mu_{ij}^t\right] + b_c\right) \\
C_{ij}^t &= f_{ij}^t \odot C_{ij}^{t-1} + z_{ij}^t \odot \tilde{C}_{ij}^t \\
h_{ij}^t &= o_{ij}^t \odot \tanh\left(C_{ij}^t\right)
\end{aligned} \tag{8}$$

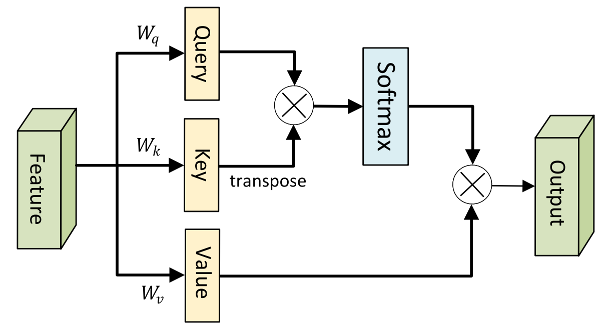

Fig. 6. Self-attention structure.

where W represents the parameter matrix, b represents the bias vector, and σ denotes the Sigmoid activation function. z^t, o^t, and f^t represent the input gate, output gate, and forget gate, respectively. Summarize the above equation as follows.

$$H_{ij} = LSTM(\boldsymbol{\mu}_{ij}) \tag{9}$$

In the above equation $\boldsymbol{\mu}_{ij}$ after one layer of LSTM to get the long-term dependencies output H_{ij}, we denote the long-term dependencies result extracted after two layers of LSTM as H'_{ij}. At the same time, the self-attention mechanism can capture not only global information but also strengthen the information weight of important times in the stock series, preserving more effective information. The calculation of attention score is as follows:

$$\begin{aligned}
Q &= H'_{ij} W_q \\
K &= H'_{ij} W_k \\
V &= H'_{ij} W_v \\
Attention(Q, K, V) &= softmax\left(\frac{QK^T}{\sqrt{d_k}} + M\right)V
\end{aligned} \tag{10}$$

where W_q, W_k, and W_v are weight matrices obtained through linear transformations, and multiplying them with the output of LSTM H'_{ij} respectively yields Q, K, and V (Fig. 6). M is a matrix with all elements above the diagonal being negative infinity and all elements on and below the diagonal being 0. The introduction of M ensures temporality, i.e., at time t, only the current time and the past information can be seen, and future information cannot be seen. $\sqrt{d_k}$ is the square root of the dimensionality d_k of the query matrix Q and key matrix K. The purpose of dividing by $\sqrt{d_k}$ is to prevent the dot product of Q and K^T from becoming too large and to avoid instability caused by large gradients during training.

Meanwhile, to reduce information loss, the output of the second LSTM layer is connected with the attention score via a residual connection to preserve more features. Finally, connect a "Dense&SiLU" module.

$$SiLU(x) = x \cdot \sigma(x) \tag{11}$$

where $\sigma(\cdot)$ is a sigmoid activation function. The SiLU activation function can calculate the gradient of the input less than 0 during backpropagation, unlike the ReLU activation function, which calculates the gradient of the input less than 0 as 0, thus avoiding the problem of gradient direction jaggedness.

3.2.2. TCNA channel

As shown in the lower part of Fig. 4, the input data also is sent into the TCNA channel for extracting local patterns and short-term dependencies features, which consist of four F-TCN blocks, a self-attention layer, and a "Dense&SiLU" module.

Based on the characteristics of the stock series, in order to ensure that the mean value and standard deviation of input and output remain

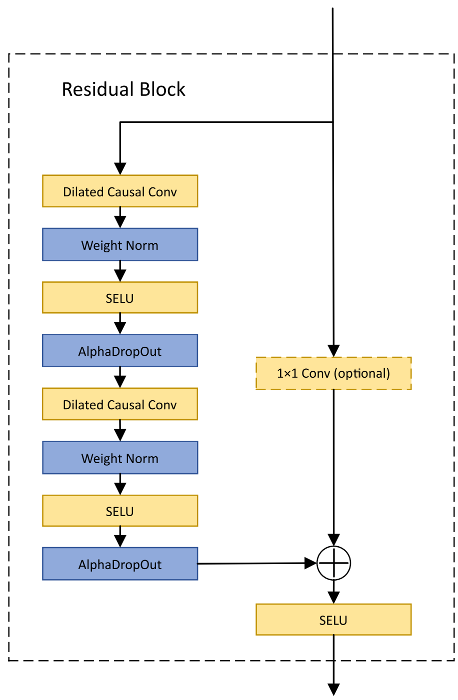

Fig. 7. F-TCN residual block.

unchanged, we construct a fine-tuning TCN block (F-TCN), as shown in Fig. 7. AlphaDropout maintains self-normalization (Klambauer et al., 2017), and the output has the same mean and standard deviation as the input. Using AlphaDropout together with the SELU activation function ensures that the output has zero mean and unit variance. Therefore, the AlphaDropout strategy is used to prevent overfitting and improve generalization capability in F-TCN, and the SELU activation function is used to speed up network convergence F-TCN. As shown in Fig. 7, the new F-TCN block is composed of dilated causal convolution layer, weight norm layer, SELU activation function, and AlphaDropout layer.

The input data is sent into a dilated causal convolution, and the output will be normalized using weight normalization. Then, the SELU activation function and AlphaDropout are applied to ensure that the output maintains the original mean and standard deviation of the input. The SELU activation function is defined as:

$$SELU(x) = \lambda \begin{cases} x, & x > 0 \\ \alpha e^x - \alpha, & x \leq 0 \end{cases} \tag{12}$$

where α is 1.673 and λ is 1.051. α and λ are defined as constants to ensure that the mean and variance of the inputs to each layer remain constant during training. This helps to mitigate the vanishing gradients problem that can occur in deep neural networks.

For the residual connection of the first F-TCN block in the TCNA channel (the deep yellow F-TCN block in Fig. 4), the number of input and output channels is different, so the input data needs to be upsampled by a 1×1 convolution before the residual connection. The other F-TCN blocks are directly connected without the 1×1 convolution. Thus n F-TCN stacks can be expressed as:

$$G_n(x) = \begin{cases} \sigma(g_1(x) + Conv_{1\times 1}(x)), & n = 1 \\ \sigma(g_n(G_{n-1}(x)) + G_{n-1}(x)), & n \geq 2 \end{cases} \tag{13}$$

where $g(\cdot)$ denotes parts such as inflated causal convolution and weight normalization (left side of Fig. 7), $Conv_{1\times 1}(\cdot)$ denotes 1×1 convolution, and σ denotes the SELU activation function (Eq. (12)), where n is 4, indicating a total of 4 F-TCN blocks.

Similar to the LSTMA channel, the attention score of the layer is calculated by Eq. (10). Finally, the output of the fourth TCN block is connected with the attention score via a residual connection and then connected to a "Dense&SiLU" module.

3.3. Prediction

Constructing specific $(LSTMA + TCNA)_j$ model for mode sub-window $\boldsymbol{\mu}_{ij}$ with specific frequency j can effectively improve prediction accuracy. The final prediction y_t can be obtained by aggregating the results of K sub-windows.

$$y_t = \sum_{j=1}^{K} \hat{y}_{t-L,j}, \quad L < t \leq N \tag{14}$$

To predict the closing price y_t on day t, it first needs predict the $\hat{y}_{t-L,j}$ based on sub-windows $\boldsymbol{\mu}_{t-L,j}, j = 1, 2, \ldots, K$, respectively. To get sub-windows $\boldsymbol{\mu}_{t-L,j}$, it needs to perform VMD on W_{t-L}. This method enables the model to effectively predict different frequency mode sub-windows separately, thereby improving prediction accuracy.

3.4. Evaluation metrics

Mean square error (MSE), root mean square error (RMSE) mean absolute error (MAE), and mean absolute percentage error (MAPE) are commonly used performance evaluation metrics in stock series prediction. Meanwhile, the R2-score is often used in regression models to assess the degree of conformity between predicted and actual values. For the first four error strategies, the smaller the error value indicates the better the prediction accuracy, while for R2, its value ranges from 0 to 1, and the closer it is to 1, the better the model fits the data. They are defined as:

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{15}$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2} \tag{16}$$

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{17}$$

$$MAPE = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \tag{18}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2} \tag{19}$$

where y_i is the real value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean value of the observed real values, and N is the length of stock series data.

4. Experimental results and analysis

This section introduces a series of evaluation metrics to evaluate the effectiveness of the new VMD-LSTMA+TCNA model on the US and Hong Kong stock markets. Firstly, compare the new VMD-LSTMA+TCNA model with other state-of-the-art deep learning models to verify the prediction accuracy. Then, the effectiveness of the decomposition strategy and the dual-channel attention hybrid network model was verified through ablation experiments. Finally, VMD was compared with other advanced decomposition methods to demonstrate that the model combination is optimal.

Table 1
Stock symbols in DJIA and HKEX.

DJIA	AAPL	AMGN	AXP	BA	CAT	CRM
	CSCO	CVX	DIS	GS	HD	HON
	IBM	INTC	JNJ	JPM	KO	MCD
	MMM	MRK	MSFT	NKE	PG	TRV
	UNH	V	VZ	WBA	WMT	
HKEX	00001	00002	00003	00005	00006	00008
	00010	00011	00012	00014	00016	00017
	00023	00027	00031	00035	00038	00041

Fig. 8. Apple stock.

4.1. Datasets

To evaluate the adaptability of our proposed model in different stock markets, we selected stock datasets from the US and Hong Kong. The list of stocks for the two datasets is shown in Table 1.

The DJIA dataset is an US stock dataset used in Liu et al. (2022). It consists of 29 US company stocks and is obtained from YahooFinance,[1] shown in Table 2. The HKEX dataset contains a total of 18 stocks from the Hong Kong stock market and is obtained from Tushare,[2] shown in Table 3.

The time range for both datasets is from January 2, 2018 to December 31, 2020, covering only the trading days. During this period, the US market had a total of 756 trading days, and the Hong Kong market had a total of 740 trading days, thus a total of 746 sub-windows were obtained on the DJIA through sliding window, and a total of 730 sub-windows are obtained on the HKEX dataset. For better understanding, we take the first stock in the DJIA dataset, AAPL (Apple), as an example, and Fig. 8 shows the daily stock price of Apple Inc. for this period, totaling 756 trading days.

To effectively validate and test the model, the datasets are split into training, validation, and test sets in a ratio of 4:1:1. To prevent data leakage, normalization is performed on mode sub-window $\boldsymbol{\mu}_{ij}$:

$$\boldsymbol{\mu}'_{ij} = \frac{\boldsymbol{\mu}_{ij} - min(\boldsymbol{\mu}_{ij})}{max(\boldsymbol{\mu}_{ij}) - min(\boldsymbol{\mu}_{ij})} \tag{20}$$

where $i = 1, 2, \dots, N - L$, $j = 1, 2, 3$. N is the data length, L is the slide window size.

At the same time, the later experiments provide only one result per dataset, which is the average calculated by all stock evaluation metrics under that dataset, so that the prediction accuracies of all stocks can be combined, which helps to provide an assessment of the overall prediction performance of the entire stock dataset and is more able to be compared intuitively.

4.2. Parameter settings

In the VMD method, we set the size L of the sliding window to 10, the moderate bandwidth constraint to 2000, the noise tolerance to 0, and the convergence criterion to 1e−7.

In the LSTMA channel, we set the size of LSTM hidden layers to 50 and the sizes of W_q, W_k, and W_v of self-attention to 50×50.

In the TCNA channel, we set the number of F-TCN residual blocks to 4, where the dilation factors of the ith block layer d is 2^{i-1}, the number of convolution channels is set to 32, and the size of the convolution kernel is 2. The dropout ratio is set to 0.2, and the 1×1 convolution is only added in the first F-TCN residual block. The size of W_q, W_k, and W_v of self-attention is 32×32.

We train the new VMD-LSTMA+TCNA model using a learning rate of 0.01 for 300 epochs and update the parameters using the Adam optimizer, and the loss function uses mean squared error (MSE).

The value of K in VMD has an impact on the predictive performance of the model. Theoretically, different optimal K values should be determined for different datasets. When it is too large, too many subsequences may lead to over-decomposition, which is difficult to interpret and analyze, and the computational complexity increases, requiring more computational resources and time. When K is too small, the decomposition results may be too coarse, leading to information loss and misinterpretation. We first analyze the decomposition efficiency under different K.

Table 4 presents the decomposition time efficiency for K set to 2, 3, 4, and 5 on both datasets. "Avg" represents the average decomposition time for individual stocks, while "Sum" indicates the total decomposition time for the entire dataset. As the value of K increases, the required decomposition time also increases. Specifically, when K is 3, the time required is four times that of K being 2; when K is 4, the time required is three times that of K being 3; and when K is 5, the time required is six times that of K being 3. It is important to note that this time refers only to the decomposition process, and a larger value of K can also lead to an extended training and inference time for the prediction model.

Meanwhile, we compare the prediction performance of VMD-LSTMA+TCNA under different K on both datasets. As shown in Table 5, the worst prediction performance on both datasets is achieved when K is 2, which is precisely because too small K leads to too coarse decomposition results, resulting in loss of information. Although the MSE metric on HKEX is at its lowest when K is 4, it is not as effective as K is 3 for the MAE, MAPE and R2-score metrics. When the decomposition number K is larger at 5, the performance tends to decrease under all metrics.

Therefore after comparing the time complexity measure and prediction performance, we selected the relatively best K as 3.

4.3. Compared F-TCN with TCN

In order to demonstrate that the F-TCN block outperforms the TCN block in prediction accuracy (Bai et al., 2018), comparative experiments were conducted on two datasets. In the models VMD-LSTMA+TCNA and VMD-LSTMA+TCN, only the TCN module has differences. We compare the prediction performance in VMD-LSTMA+TCNA using F-TCN and original TCN.

As shown in Table 6, on the DJIA dataset, the F-TCN outperforms the TCN in terms of MSE and RMSE while performing similarly in terms of MAE, MAPE and R2 score. On the HKEX dataset, the F-TCN outperforms the TCN in all metrics.

[1] https://finance.yahoo.com/.
[2] https://tushare.pro/.

Table 2
DJIA dataset describe.

Stock code	Name	Total	Total window	Training	Validate	Test
AAPL	Apple	756	746	500	124	122
AMGN	Amgen	756	746	500	124	122
…	…	…	…	…	…	…
WBA	Walgreens Boots Alliance	756	746	500	124	122
WMT	Walmart	756	746	500	124	122

Table 3
HKEX dataset describe.

Stock code	Name	Total	Total window	Training	Validate	Test
00001.HK	CK Hutchison Holdings Limited	740	730	489	122	119
00002.HK	CLP Holdings Limited	740	730	489	122	119
…	…	…	…	…	…	…
00038.HK	First Tractor	740	730	489	122	119
00041.HK	Great Eagle Holdings Limited	740	730	489	122	119

Table 4
Time required for decomposition of different K.

	Time (s)	2	3	4	5
DJIA	Avg	1.07	3.98	12.37	22.96
	Sum	31.30	115.50	358.74	666.12
HKEX	Avg	0.90	3.69	9.67	16.70
	Sum	16.29	66.49	174.06	300.68

Table 5
Performance of VMD-LSTMA+TCNA at different K.

K	DJIA					HKEX				
	MSE	RMSE	MAE	MAPE	R2	MSE	RMSE	MAE	MAPE	R2
2	7.26	2.32	1.64	1.13	0.92	0.42	0.48	0.35	1.09	0.91
3	**6.15**	**2.14**	**1.48**	**1.01**	**0.93**	0.38	**0.45**	**0.31**	**0.96**	**0.93**
4	6.93	2.42	1.53	1.03	0.92	**0.37**	**0.45**	0.32	0.99	0.92
5	7.36	2.30	1.60	1.09	0.92	0.38	0.47	0.34	1.06	0.91

Table 6
Compare F-TCN with TCN.

Dataset	Model	MSE	RMSE	MAE	MAPE	R2
DJIA	TCN	6.37	2.16	**1.48**	**1.01**	**0.93**
	F-TCN	**6.15**	**2.14**	**1.48**	**1.01**	**0.93**
HKEX	TCN	0.41	0.47	0.33	1.01	0.92
	F-TCN	**0.38**	**0.45**	**0.31**	**0.96**	**0.93**

Table 7
Prediction performance of seven methods on DJIA.

Model	MSE	RMSE	MAE	MAPE	R2
GRU	10.47	2.75	1.97	1.33	0.89
LSTM	10.42	2.75	1.96	1.34	0.89
N-BEATS	10.96	2.81	2.01	1.37	0.88
CNN-LSTM	10.74	2.77	1.98	1.35	0.89
PF-LSTM	10.36	2.74	1.96	1.33	0.89
VML	7.05	2.26	1.55	1.05	0.92
VMD-LSTMA+TCNA	**6.15**	**2.14**	**1.48**	**1.01**	**0.93**

Table 8
Prediction performance of seven methods on HKEX.

Model	MSE	RMSE	MAE	MAPE	R2
GRU	0.60	0.57	0.42	1.29	0.88
LSTM	0.62	0.58	0.42	1.29	0.87
N-BEATS	0.61	0.58	0.42	1.32	0.88
CNN-LSTM	0.59	0.58	0.42	1.29	0.88
PF-LSTM	0.58	0.57	0.42	1.29	0.88
VML	0.44	0.48	0.34	1.05	0.92
VMD-LSTMA+TCNA	**0.38**	**0.45**	**0.31**	**0.96**	**0.93**

Table 9
Average and total inference time for several methods with two datasets.

Dataset	DJIA		HKEX	
time (s)	Avg	Sum	Avg	Sum
N-Beats	0.20	5.79	0.22	3.99
GRU	**0.16**	**4.66**	**0.20**	**3.63**
LSTM	**0.16**	4.80	0.22	3.93
CNN-LSTM	0.22	6.47	0.29	5.30
PF-LSTM	0.47	13.53	0.43	7.73
VML	2.50	72.58	2.46	44.22
VMD-LSTMA + TCNA	0.34	9.88	0.42	7.58

4.4. Prediction performance

To demonstrate the stock series prediction performance of VMD-LSTMA+TCNA, We compared it with other state-of-the-art methods. N-BEATS pioneers a new backbone for time series forecasting, which is realized only through full connectivity (Oreshkin et al., 2019), which we use for stock series forecasting. CNN-LSTM is a combination of CNN and LSTM that aims to improve the stock series prediction accuracy by leveraging the advantages of CNN in extracting effective features from data and LSTM in discovering the dependencies of data in stock series (Lu et al., 2020). PF-LSTM is a combination of LSTM and particle filtering for text categorization and stock sequence prediction (Ma et al., 2020). VML is a new stock series prediction method that combines VMD to make the stock series smooth and the Model-Agnostic Meta-Learning algorithm (MAML) with LSTM to enable the model to fine-tune according to the latest stock data and constantly adapt to new data (Liu et al., 2022). In addition, two baseline models LSTM and GRU are used for comparison as well.

As shown in Tables 7 and 8, we can see that VMD-LSTMA+TCNA outperforms all of them in terms of all evaluation metrics, proving the effectiveness of our proposed method.

N-BEATS, while showing excellent performance in time series forecasting, performs poorly in stock series forecasting. Meanwhile, although the state-of-the-art method VML improves the forecasting performance by obtaining excellent initialization parameters of LSTM through meta-learning algorithms, it is still weaker than our method precisely because VML only focuses on the long-term dependencies while ignoring the local features and short-term dependencies.

We also plotted the cumulative MAE curves of these models on the two datasets, using CRM in the US stock market and 00002.HK in the Hong Kong stock market. As shown in Fig. 9, we can observe that the error curves of CNN-LSTM are higher than those of the hybrid prediction methods based on decomposition, indicating the effectiveness of the decomposition strategy. The error curve of our proposed model remains at the bottom with the lowest error, which proves the effectiveness of our method.

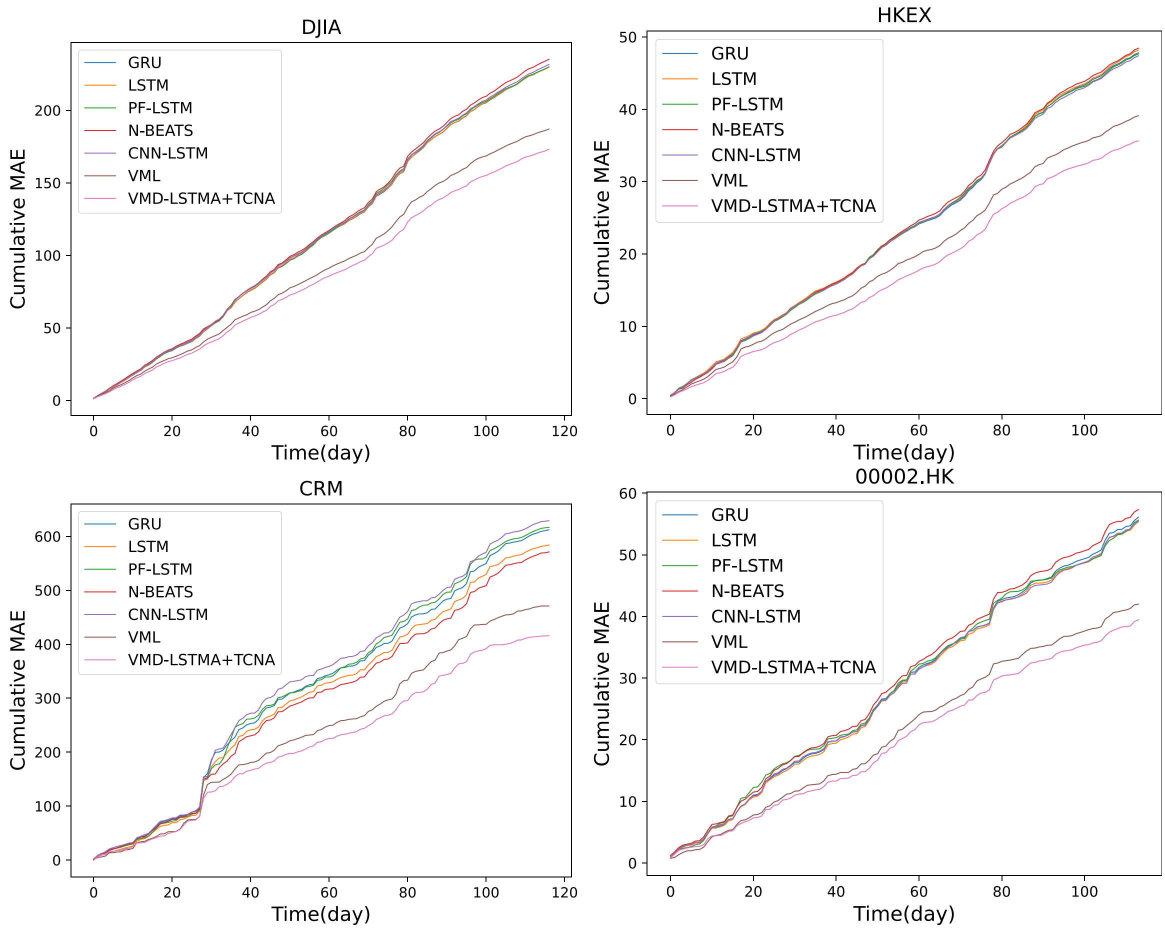

Fig. 9. Prediction performance of five methods.

Additionally, we evaluated the inference time of several models on two datasets (Table 9). The LSTM and GRU baseline models have relatively faster inference times, but their predictive performance is inferior. VML has the longest inference time due to the time-consuming nature of the meta-learning algorithm MAML (Model-Agnostic Meta-Learning) in task generation and finding optimal parameters across multiple tasks. However, our proposed VMD-LSTMA+TCNA model's inference time is better than that of PF-LSTM and VML.

In addition, we compared the floating point operations (FLOPs) and model parameter sizes of several models (Fig. 10). The PF-LSTM model has the highest FLOPs, followed by VMD-LSTMA+TCNA. Our proposed VMD-LSTMA+TCNA model has the most significant number of parameters among the models, enabling it to extract more information from stock series and achieve the best predictive performance in the stock market.

4.5. Ablation study

In order to demonstrate the effectiveness of each module in our proposed method, we conducted the following ablation experiments: The first is to use VMD and LSTM prediction; The second is to use VMD and LSTM with self-attention for prediction; The third is to use VMD and TCN to prediction; The fourth is to use VMD to decompose and use TCN with self-attention to prediction; The fifth is to use VMD to decompose and then combine TCN and LSTM to prediction; The sixth is to directly use the combination of TCN with self-attention and LSTM for prediction; The last one is our proposed VMD-LSTMA+TCNA. These experiments aimed to verify the contribution of each module in our proposed model to the prediction performance.

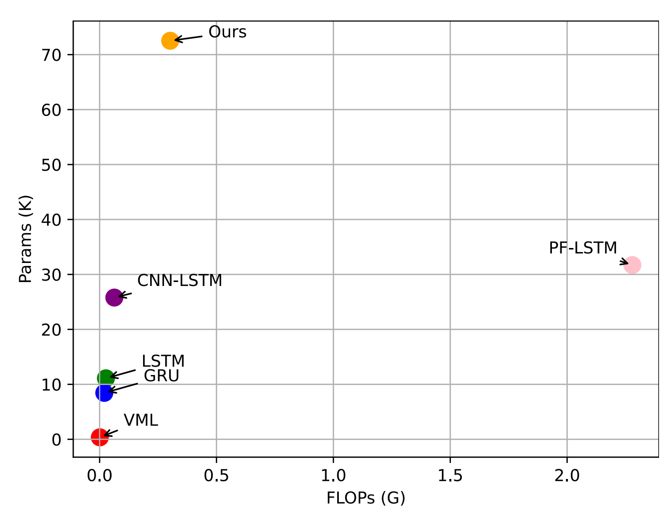

Fig. 10. FLOPs and Params for several methods, the x-axis indicates the size of FLOPs, and the y-axis indicates the size of the model parameters.

Tables 10 and 11 quantitatively present the MSE, RMSE, MAE and MAPE of the seven models on the DJIA and HKEX datasets. The experimental results of {1, 2} and {3, 4} indicate that adding self-attention is better than not, implying that self-attention can leverage the stock series information at important times. {1, 3, 5} experiments have shown that the dual-channel structure is able to capture both long-term and short-term dependencies features and has better predictive

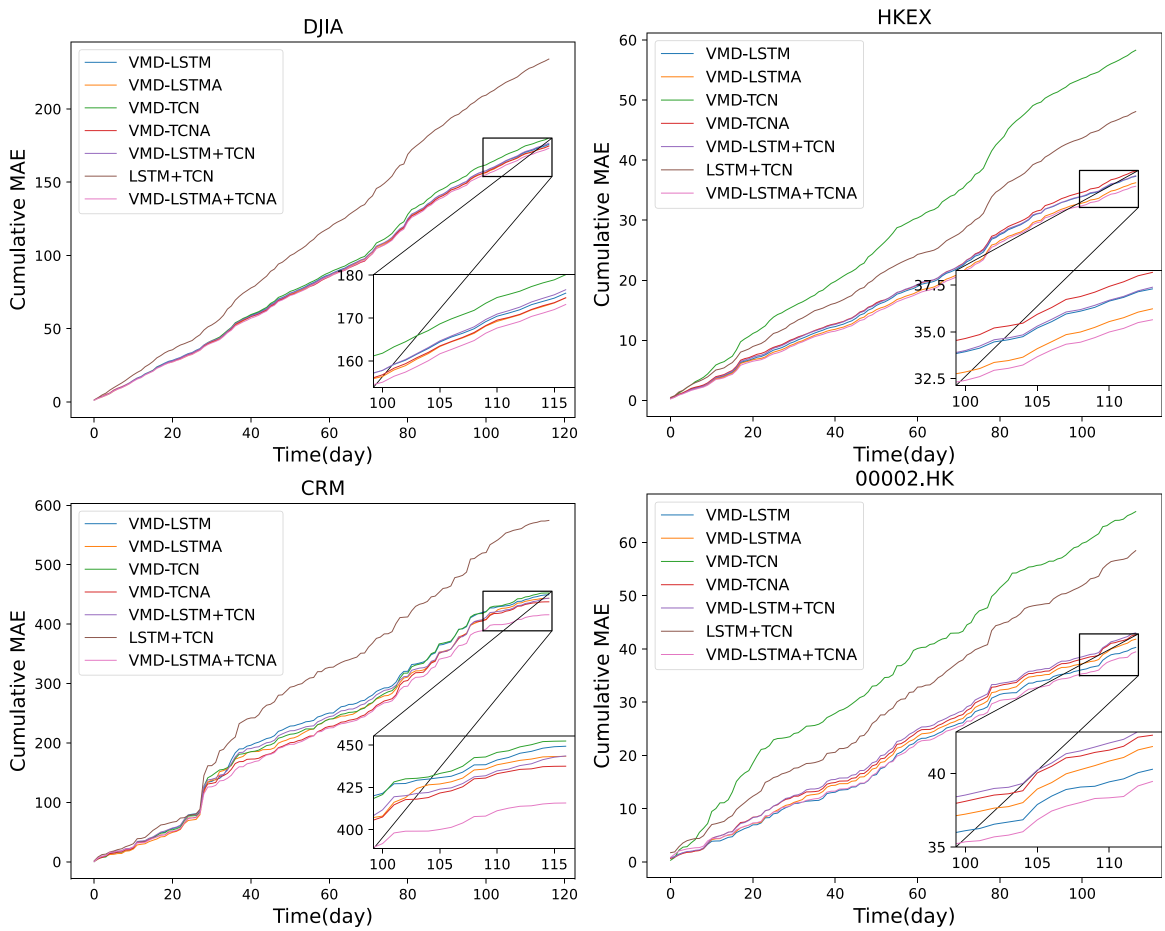

Fig. 11. Prediction performance of seven models.

Table 10
Prediction performance on DJIA.

Model	MSE	RMSE	MAE	MAPE	R2
1. VMD-LSTM	6.56	2.18	1.50	1.02	**0.93**
2. VMD-LSTMA	6.42	2.16	1.49	1.02	**0.93**
3. VMD-TCN	6.87	2.23	1.54	1.05	0.92
4. VMD-TCNA	6.35	2.16	1.49	1.02	**0.93**
5. VMD-LSTM + TCN	6.52	2.18	1.51	1.02	**0.93**
6. LSTMA + TCNA	10.37	2.76	2.00	1.37	0.88
7. VMD-LSTMA + TCNA	**6.15**	**2.14**	**1.48**	**1.01**	**0.93**

Table 11
Prediction performance on HKEX.

Model	MSE	RMSE	MAE	MAPE	R2
1. VMD-LSTM	0.41	0.46	0.33	1.00	0.92
2. VMD-LSTMA	0.39	0.46	0.32	**0.96**	**0.93**
3. VMD-TCN	0.77	0.67	0.49	1.57	0.81
4. VMD-TCNA	0.41	0.47	0.33	1.03	0.92
5. VMD-LSTM + TCN	0.40	0.46	0.33	1.00	0.92
6. LSTMA + TCNA	0.58	0.58	0.42	1.30	0.88
7. VMD-LSTMA + TCNA	**0.38**	**0.45**	**0.31**	**0.96**	**0.93**

performance than a model that captures a single long-term or short-term feature. The experimental results of {6, 7} show that using the decomposition strategy can have a greater effect than not using it. Overall, our proposed model has achieved the best results in all evaluation metrics.

The cumulative error curves plotted in Fig. 11 demonstrate that our proposed model consistently outperforms the other models on both datasets. Over time, the curve of VMD-LSTMA+TCNA remains at the bottom and gradually pulls away from the other curves, indicating that our model has better long-term prediction performance.

Table 12
Different decomposition methods prediction performance on DJIA.

Model	MSE	RMSE	MAE	MAPE	R2
DWT-LSTMA + TCNA	6.45	2.18	1.50	**1.01**	**0.93**
EMD-LSTMA + TCNA	8.26	2.57	1.89	1.40	0.85
EEMD-LSTMA + TCNA	7.49	2.41	1.80	1.24	0.90
CEEMDAN-LSTMA + TCNA	9.05	2.62	1.94	1.35	0.89
VMD-LSTMA + TCNA	**6.15**	**2.14**	**1.48**	**1.01**	**0.93**

4.6. Compared with other decomposition methods

Next, to demonstrate that VMD is the most suitable decomposition method for our model, we replace the VMD decomposition strategy in VMD-LSTMA+TCNA with other decomposition methods, including discrete wavelet transform (DWT), empirical mode decomposition (EMD) (Huang et al., 1998), improved ensemble empirical mode decomposition (EEMD) (Wu & Huang, 2009), and complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) (Torres et al., 2011).

The experimental results are shown in Tables 12 and 13. Compared with EMD, the improved EEMD method shows better performance on both datasets. Our proposed model performs best on the DJIA dataset, followed by wavelet decomposition, but the prediction performance of wavelet decomposition on the HKEX dataset is not as good as that of EEMD and CEEMDAN. The improved CEEMDAN does not perform as well as EMD on the DJIA dataset, indicating that its generalization

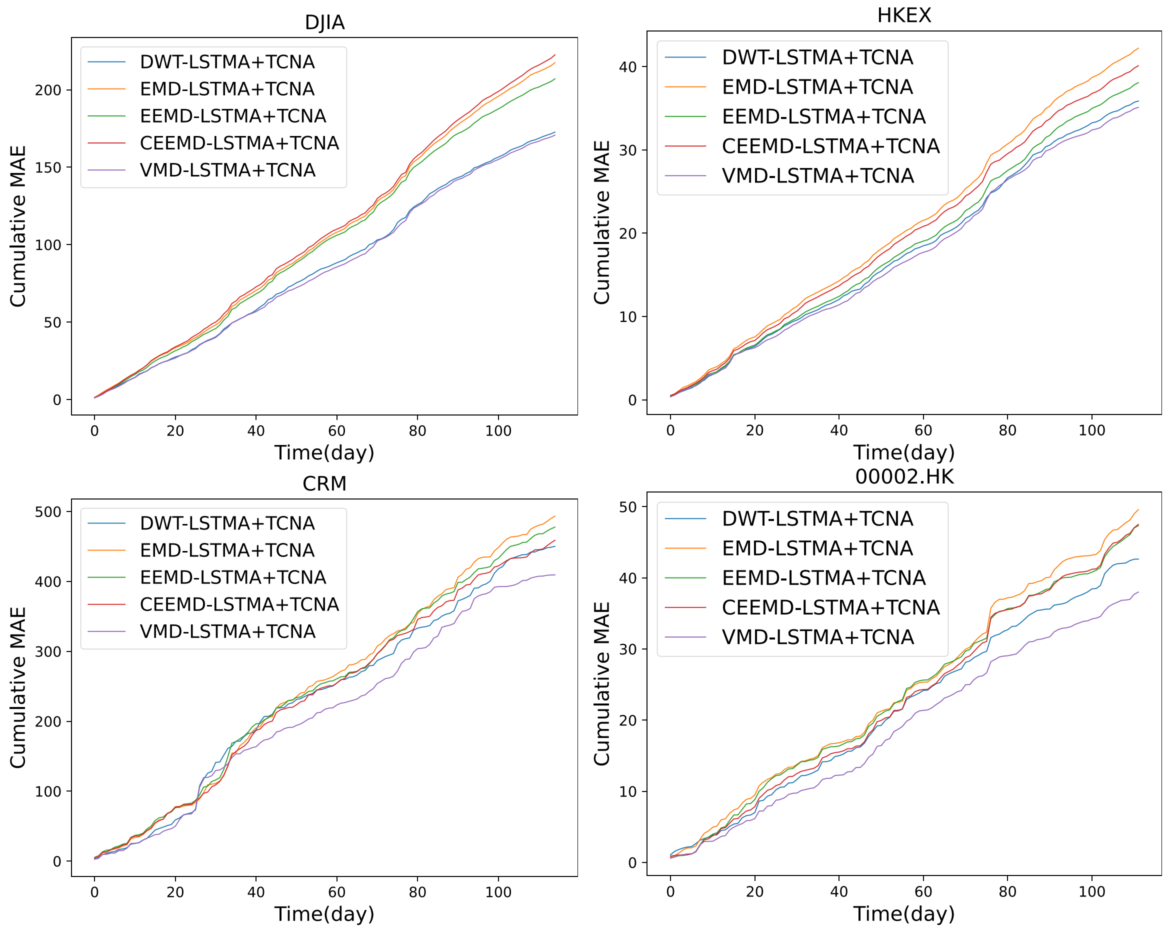

Fig. 12. Prediction performance of different decomposition methods.

Table 13
Different decomposition methods prediction performance on HKEX.

Model	MSE	RMSE	MAE	MAPE	R2
DWT-LSTMA + TCNA	0.41	0.46	0.32	0.97	0.92
EMD-LSTMA + TCNA	0.44	0.51	0.38	1.70	0.90
EEMD-LSTMA + TCNA	0.39	0.46	0.34	1.10	0.91
CEEMDAN-LSTMA + TCNA	0.40	0.48	0.36	1.14	0.91
VMD-LSTMA + TCNA	**0.38**	**0.45**	**0.31**	**0.96**	**0.93**

Table 14
Traffic flow prediction comparison.

Model	MSE	RMSE	MAE	MAPE	R2
LSTM	98.05	9.90	7.21	**16.56**	0.94
GRU	99.32	9.97	7.20	16.78	0.94
SAEs	92.08	9.60	7.06	17.80	0.94
VML	73.69	8.58	6.51	17.70	**0.96**
VMD-LSTMA + TCNA	**66.05**	**8.13**	**5.98**	17.54	**0.96**

ability to adapt to different datasets is not strong enough. On both datasets, only our proposed model achieves the best scores on all evaluation metrics, indicating that the performance of VMD is superior to other decomposition methods.

Based on the comparison in Fig. 12, it can be seen that the hybrid algorithm based on VMD has the best performance, with the cumulative error consistently at the bottom. The combination of DWT also performs well and has a relatively low cumulative error.

4.7. Applied for predictions in other domains

4.7.1. Traffic flow prediction

To confirm the generalization and effectiveness of VMD-LSTMA+TCNA, we applied it to traffic flow prediction and compared its performance with LSTM, GRU, SAEs (Lv et al., 2014) and VML (Liu et al., 2022). The PeMS 5-min interval traffic flow data is used, which records traffic flow every five minutes.

The experiment results in Table 14 show that VMD-LSTMA+TCNA is also effective in traffic flow prediction. VMD-LSTMA+TCNA achieves the best performance in the evaluation metrics of MSE, RMSE, MAE and R2 score. However, it performs poorly on MAPE due to the non-linear nature and the difficulty in accurately predicting extreme values. It is also worth noting that VML achieves better performance than LSTM, GRU and SAES in all evaluation metrics, indicating the effectiveness of the decomposition strategy in improving the generalization ability of the model.

4.7.2. Wind speed prediction

We have also extended VMD-LSTMA+TCNA to wind speed prediction, using the same Inner Mongolia wind speed dataset as in Yan et al. (2020). We also referred to some experimental data from Liu et al. (2022) and Yan et al. (2020).

As shown in Table 15, due to some of the relevant methods have yet to be open-sourced, in order to ensure a fair comparison, we rely on the experimental results provided by the corresponding papers. For cases where the paper did not provide results, we use "∼" to indicate this. It can be seen that VMD-TCNA-LSTMA is generally effective and performs better than VML with the same decomposition strategy. However, ISSD-LSTM-GOADBN had the best performance, but its decomposition object is the entire time series, leading to data leakage.

Table 15
Wind speed prediction comparison.

Model	MSE	RMSE	MAE	MAPE	R2
LSTM	0.60	0.77	0.63	14.56	0.85
ISSD-BP	0.49	0.70	0.54	12.79	~
ISSD-SVM	0.40	0.63	0.46	10.17	~
ISSD-LSTM	0.30	0.55	0.45	9.35	~
VML	0.24	0.49	0.37	8.79	0.96
VMD-LSTM-GOADBN	0.24	0.49	0.35	7.19	~
VMD-LSTMA + TCNA	0.22	0.46	0.36	8.52	0.96
SSD-LSTM-GOADBN	0.21	0.46	0.34	7.22	~
ISSD-LSTM-GOADBN	**0.08**	**0.29**	**0.12**	**2.56**	~

Table 16
Disease spread prediction comparison.

Model	MSE	RMSE	MAE	MAPE	R2
LSTM	1.84×10^6	1355.68	616.63	20.01	0.92
GRU	1.71×10^6	1307.74	588.86	19.49	0.92
VMD-LSTM	1.63×10^6	1276.71	541.12	**14.54**	0.92
VMD-GRU	1.66×10^6	1288.74	550.04	15.61	0.92
VMD-LSTMA	1.47×10^6	1214.39	507.05	15.08	0.93
VML	2.20×10^6	1484.65	665.47	18.62	0.90
VMD-LSTMA + TCNA	1.17×10^6	**1080.29**	**501.25**	17.96	**0.95**

4.7.3. Disease spread prediction

To provide a more comprehensive demonstration of the applicability of VMD-TCNA+LSTMA in various domains, we extended its application to the field of disease spread prediction. We utilized the Influenza-Like Illness (ILI) public dataset, which can be accessed from the link.[3] This dataset comprises weekly reports from the U.S. Centers for Disease Control and Prevention (CDC) spanning from 2002 to 2021, depicting the ratio of influenza-like illness patients to total patients within a week. The dataset encompasses 996 data entries and 7 variables. We specifically employed the "Number of Providers" (NUM. OF PROVIDERS) variable from the ILI dataset for our experiments. The methods for comparison have been introduced in the preceding experiments.

The comparison of prediction results among several methods on the ILI dataset is presented in Table 16. Given that the ILI dataset records thousands of new influenza cases each week, resulting in larger MSE values (scientific notation is employed in the table). VMD-LSTMA+TCNA outperforms other methods in four out of five evaluation metrics, and this indicates the effectiveness of the proposed approach. Additionally, the performance of the meta-learning-based VML method (Liu et al., 2022) is subpar on this dataset, coupled with slower training times. Various models utilizing VMD demonstrate relative performance enhancements, with VMD-LSTMA exhibiting further improvement due to the incorporation of the self-attention mechanism. In summary, these experiments substantiate the versatility and applicability of the proposed method outlined in this paper.

5. Conclusion

In this paper, we proposed a stock series prediction model based on variational mode decomposition and dual-channel self-attention network (VMD-LSTMA+TCNA). Firstly, the stock series is divided into different frequency mode sub-windows using a sliding window and VMD. Then, we use the dual-channel attention network with self-attention to predict the mode sub-window series. Our proposed model achieves higher prediction accuracy compared to other latest methods.

In future research, it is necessary to delve deeper into the characteristics of the stock market to make the model more realistic and expand its applicability to obtain better prediction accuracy. We also hope that the proposed model can be applied to other scenarios, such as traffic flow and wind power prediction.

[3] https://gis.cdc.gov/grasp/fluview/fluportaldashboard.html.

CRediT authorship contribution statement

Yepeng Liu: Conceptualization, Methodology, Software, Writing – original draft, Visualization, Writing – review & editing. **Siyuan Huang:** Conceptualization, Methodology, Software, Writing – original draft, Visualization, Writing – review & editing. **Xiaoyi Tian:** Writing – review & editing. **Fan Zhang:** Writing – review & editing, Funding acquisition. **Feng Zhao:** Writing – original draft, Funding acquisition. **Caiming Zhang:** Conceptualization, Resources, Validation, Funding acquisition, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported partly by the National Natural Science Foundation of China (Grant No. 62272281, No. 62002200, No. 62202268). Shandong Provincial Natural Science Foundation (Grant No. ZR2021QF134, No. ZR2020QF012). Shandong Provincial Science and Technology Support Program of Youth Innovation Team in Colleges (Grant No. 2021KJ069). Shandong technology and business university wealth management interdisciplinary research project (2023TSXM008).

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